

Slope or y -Intercept? Find the Mix-Up

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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| 1. (a) slope = 12, (b) y-intercept = 40 | 4, — |
| 2. (a) slope = 3, (b) y when x is 2 = 10, — | 6. (a) height after 4 hours = 8, — |
| 3. (a) y-intercept = 2, (b) rise per unit = 7, — | 7. (a) slopes compared = $>$, (b) y-intercepts compared = $<$, — |
| 4. (a) dollars per gigabyte = 4, (b) cost at 8 gigabytes = 57, — | 8. (a) dollars per week = 9, (b) balance at week 10 = 150, — |
| 5. (a) cost at 0 miles = 3, (b) dollars per mile = | |
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What is verified

1 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.B.6 – problems 2, 4, 8

8.F.B – problems 1, 3, 5, 6, 7

Difficulty 1–4 of 5 across 22 tagged checks.

Common wrong answers

2. If they got 11: swapped the slope and the y -intercept, then substituted
4. If they got 204: swapped the rate and the starting cost, then substituted
6. If they got -77 : swapped the starting height and the burn rate, then substituted
8. If they got 609: swapped the weekly rate and the starting balance, then substituted
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1. $C = 12x + 40$ dollars for x months. (a) Slope. (b) y -intercept.

Solution (a): One more month changes the cost from $12x + 40$ to $12(x + 1) + 40$, a difference of 12 dollars. That is the number multiplying x .

12

Solution (b): Substitute $x = 0$: the $12x$ term vanishes and 40 is left.

40

2. Line through $(0, 4)$ and $(5, 19)$; Tomas wrote $y = 4x + 3$. (a) Slope. (b) y at $x = 2$ from the correct equation. (c) The error.

Solution (a): $\frac{19 - 4}{5 - 0} = \frac{15}{5}.$

$m = 3$

Solution (b): The point $(0, 4)$ is already on the y -axis, so the y -intercept is 4 and the correct equation is $y = 3x + 4$. At $x = 2$: $3(2) + 4$.

$y = 10$

Solution (c) — instructor-judged. The two numbers are in each other's places. Because 4 is the value of y when x is 0, it is the y -intercept; 3 is the amount y grows for each unit of x , so it is the slope. His equation would give $y = 11$ at $x = 2$, one more than the true value — a small gap here, but the same swap is catastrophic in problems 4 and 8. A correct response names which number belongs where.

3. Nia says $y = 7x + 2$ crosses the y -axis at 7 and rises 2 per unit. (a) Crossing value. (b) Rise per unit. (c) The error.

Solution (a): Substitute $x = 0$: $7(0) + 2$.

2

Solution (b): Going from x to $x + 1$ adds 7 to y .

7

Solution (c) — instructor-judged. She has the two roles reversed. The number multiplying x is the rise per unit across; the number added on is the height where the graph meets the vertical axis. Substituting $x = 0$ is the quickest way to show it. A pair of corrected numbers with no statement of which role each plays does not earn full credit.

4. Phone plan: \$25 at 0 GB and \$45 at 5 GB; Reza wrote $C = 25g + 4$. (a) Cost per gigabyte. (b) Cost at 8 GB from the correct equation. (c) The error.

Solution (a): $\frac{45 - 25}{5 - 0} = \frac{20}{5}.$

$m = 4$

Solution (b): The cost at 0 GB is 25, so the correct equation is $C = 4g + 25$. At $g = 8$: $4(8) + 25$.

57

Solution (c) — instructor-judged. 25 dollars is what the plan costs before any data is used, so it is the constant; 4 dollars is the extra cost per gigabyte, so it multiplies g . His equation predicts 204 dollars for 8 gigabytes, which is more than four times the true cost. A correct response names which number belongs where; testing the equation at 0 GB is valid evidence.

5. Taxi table: miles 0, 2, 4, 6 against charges \$3, \$11, \$19, \$27. Kai says $y = 3x + 4$. (a) Charge at 0 miles. (b) Charge per mile. (c) The error.

Solution (a): Read the first column of the table.

3

Solution (b): From the first column to the last, the charge rises $27 - 3 = 24$ dollars over $6 - 0 = 6$ miles, so $\frac{24}{6}$.

$m = 4$

Solution (c) — instructor-judged. The table's first row gives the charge at zero miles, which belongs as the constant; the step from row to row gives the charge per mile, which multiplies the variable. Kai has them the wrong way round, and his equation fails at every row of the table — at 2 miles it gives 10 dollars, not 11. A correct response names the roles; a corrected equation on its own does not.

6. Candle: $h = 20 - 3t$ centimetres; Lena wrote $h = 3 - 20t$. (a) Height after 4 hours. (b) The error.

Solution (a): $20 - 3(4) = 20 - 12$.

8

Solution (b) — instructor-judged. 20 cm is the height before any burning, so it is the constant; 3 cm per hour is the loss rate, so it multiplies t . Her version gives $3 - 20(4) = -77$ cm after four hours, and a candle cannot have a negative height — valid supporting evidence, but the roles of the two numbers must be named for full credit.

7. $y = 5x + 2$ and $y = 2x + 5$. (a) Compare slopes. (b) Compare y -intercepts. (c) Why the lines differ.

Solution (a): The slopes are 5 and 2, first line then second.

$$\boxed{5 > 2}$$

Solution (b): The y -intercepts are 2 and 5, in the same order.

$$\boxed{2 < 5}$$

Solution (c) — instructor-judged. Position decides the role. The number multiplying x sets the steepness and the number added on sets where the line starts, so exchanging them exchanges a steep line starting low for a shallow line starting high. The two graphs cross at $(1, 7)$ and go different ways from there. “The numbers are in different places” with no account of what each place means is not enough.

8. Savings: \$60 at week 0 and \$132 at week 8; Marcus wrote $y = 60x + 9$. (a) Amount saved per week. (b) Balance at week 10. (c) Why his prediction is far too large.

Solution (a): $\frac{132 - 60}{8 - 0} = \frac{72}{8}$.

$$\boxed{m = 9}$$

Solution (b): The balance at week 0 is 60, so the correct equation is $y = 9x + 60$. At $x = 10$: $9(10) + 60$.

$$\boxed{150}$$

Solution (c) — instructor-judged. An estimate settles it without computing his value: the account gained 72 dollars in the first eight weeks, so two more weeks add only about twenty more — nothing near the hundreds. His equation would give 609 dollars, which is roughly four times the true balance. Accept any rough argument of this shape; a full recomputation presented as the whole answer misses the point of the question.